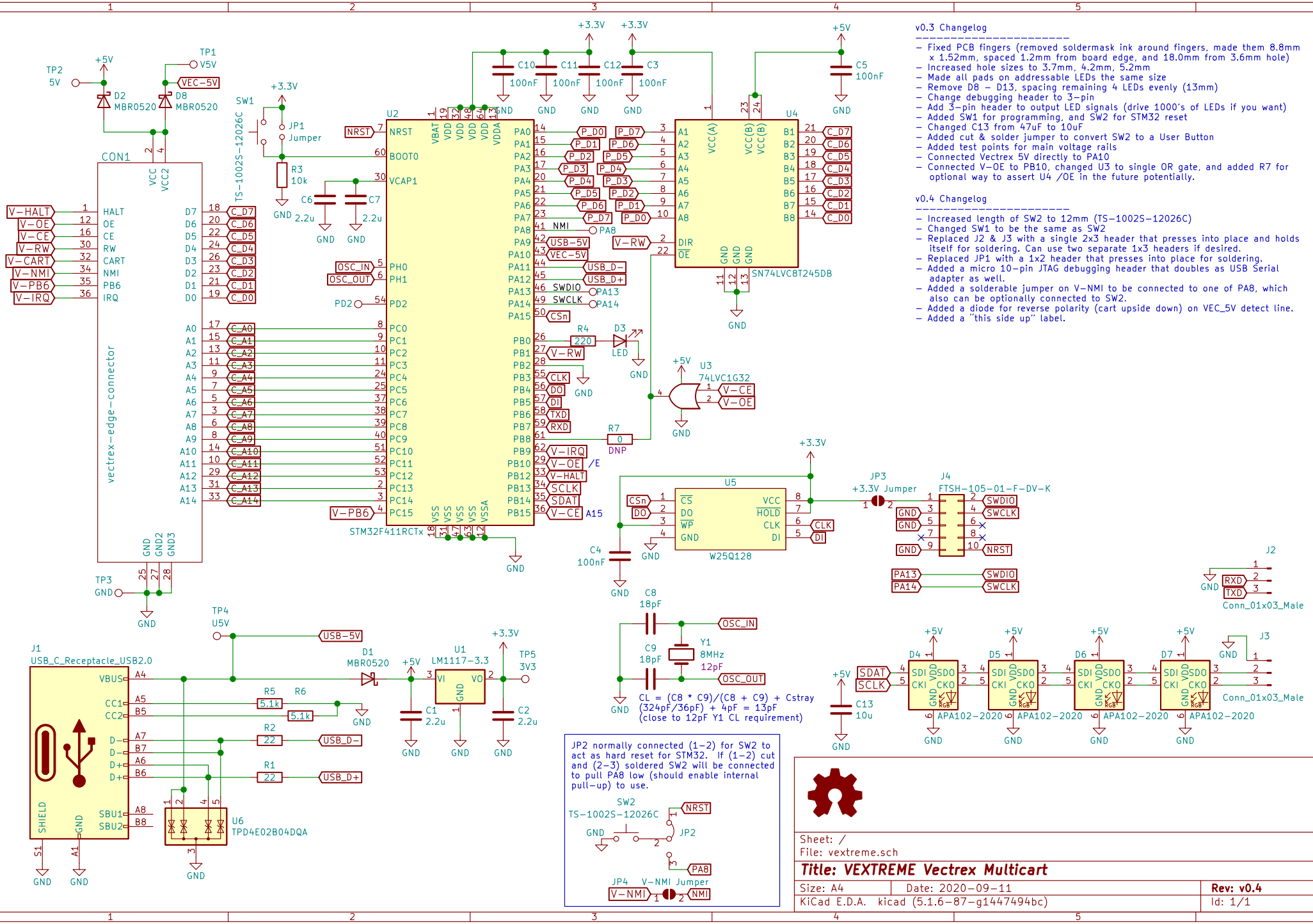

v0.3 Changelog
– Fixed PCB fingers (removed soldermask ink around fingers, made them 8.8mm x 1.52mm, spaced 1.2mm from board edge, and 18.0mm from 3.6mm hole)
– Increased hole sizes to 3.7mm, 4.2mm, 5.2mm
– Made all pads on addressable LEDs the same size
– Remove D8 – D13, spacing remaining 4 LEDs evenly (13mm)
– Change debugging header to 3-pin
– Add 3-pin header to output LED signals (drive 1000's of LEDs if you want)
– Added SW1 for programming, and SW2 for STM32 reset
– Changed C13 from 47uF to 10uF
– Added cut & solder jumper to convert SW2 to a User Button
– Added test points for main voltage rails
– Connected Vectrex 5V directly to PA10
– Connected V–OE to PB10, changed U3 to single OR gate, and added R7 for optional way to assert U4 /OE in the future potentially.
v0.4 Changelog
– Increased length of SW2 to 12mm (TS–1002S–12026C)
– Changed SW1 to be the same as SW2
– Replaced J2 & J3 with a single 2x3 header that presses into place and holds itself for soldering. Can use two separate 1x3 headers if desired.
– Replaced JP1 with a 1x2 header that presses into place for soldering.
– Added a micro 10–pin JTAG debugging header that doubles as USB Serial adapter as well.
– Added a solderable jumper on V–NMI to be connected to one of PA8, which also can be optionally connected to SW2.
– Added a diode for reverse polarity (cart upside down) on VEC_5V detect line.
– Added a "this side up" label.
CON1
vectrex–edge–connector
U2
STM32F411RCTx
U4
SN74LVC8T245DB
U3
74LVC1G32
U5
W25Q128
U1
LM1117–3.3
U6
TPD4E02B04DQA
J1
USB_C_Receptacle_USB2.0
SW1
TS–1002S–12026C
SW2
TS–1002S–12026C
JP1
Jumper
JP2
JP3
+3.3V Jumper
JP4
V–NMI Jumper
J2
J3
J4
FTSH–105–01–F–DV–K
D1
MBR0520
D2
MBR0520
D8
MBR0520
D3
LED
D4
D5
D6
D7
APA102–2020
Y1
8MHz
12pF
CL = (C8 * C9)/(C8 + C9) + Cstray
(324pF/36pF) + 4pF = 13pF
(close to 12pF Y1 CL requirement)
JP2 normally connected (1–2) for SW2 to act as hard reset for STM32. If (1–2) cut and (2–3) soldered SW2 will be connected to pull PA8 low (should enable internal pull–up) to use.
Sheet: /
File: vextreme.sch
Title: VEXTREME Vectrex Multicart
Size: A4
Date: 2020–09–11
Rev: v0.4
KiCad E.D.A. kicad (5.1.6–87–g1447494bc)
Id: 1/1